

Characterization of conserved Flavivirus 5'UTR elements

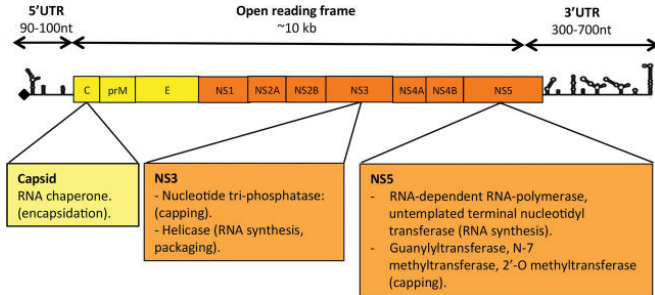
Michael T. Wolfinger

Department of Theoretical Chemistry
University of Vienna

Bled, Slovenia, February 16, 2017

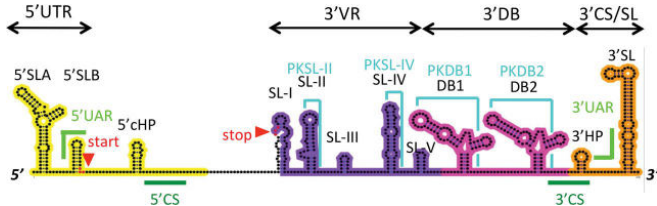
Flavivirus genome characteristics

- Single-stranded, positive-strand RNA viruses
- Enters cell through receptor-mediated endocytosis
- Capped, non-polyadenylated genome (gRNA) of 10-12kb length
- Encodes a single ORF, flanked by structured 5'-UTR and 3'-UTR
- Translation of FV ORF yields a single polyprotein



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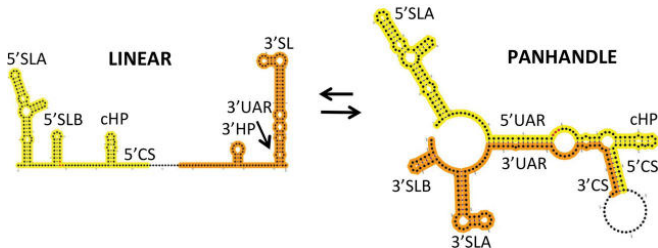


Flavivirus replication

- FV RNA replication happens via asymmetric replication cycle
- Viral RNA-dependent RNA polymerase (RdRp) synthesizes a full-length minus-strand (-)gRNA
- (-)gRNA is a template for several rounds of (+)gRNA synthesis
- Ratio (-)gRNA / (+)gRNA between 1:10 and 1:100
- Genome cyclisation is required for (-)gRNA synthesis

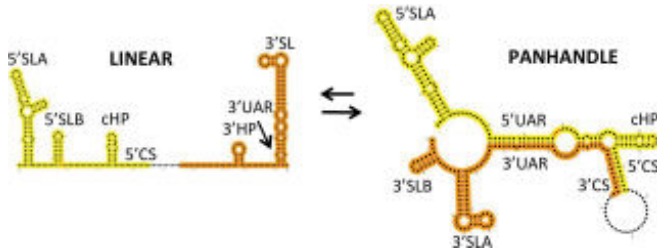
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Flavivirus replication

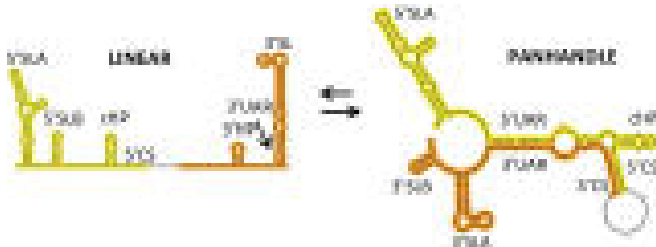
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- SLA acts as promoter / enhancer for viral RdRp

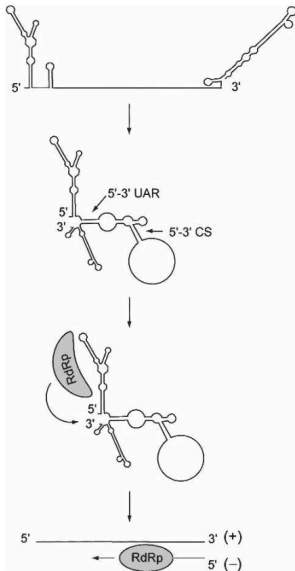
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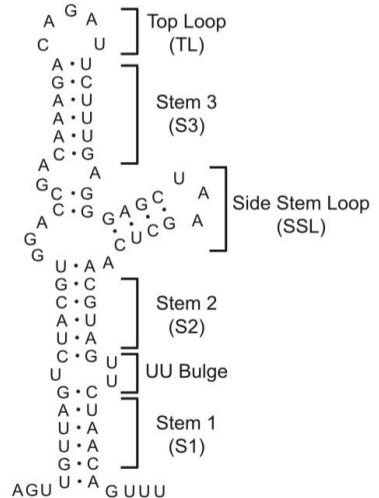
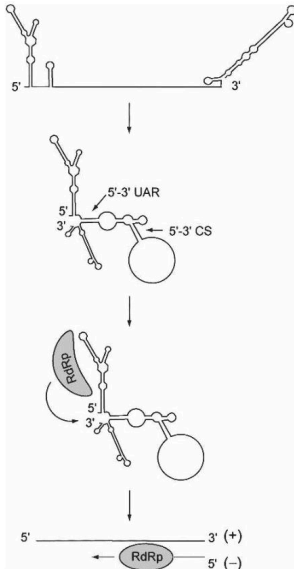
- SLA acts as promoter / enhancer for viral RdRp
- Viral RdRps are **error-prone**: Misincorporations every 40-70kb

Flavivirus replication



Lodeiro et al., J. Virol. (2009)

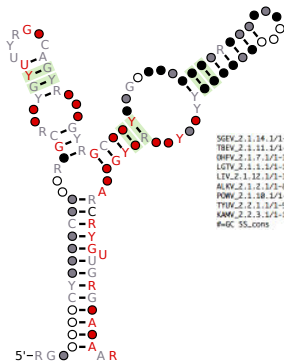
Flavivirus replication



SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

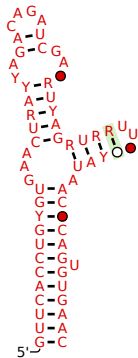
Tick-borne Flavivirus group (TBFVG)

[illegible]

SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

Kokobera virus group (KOKVG)



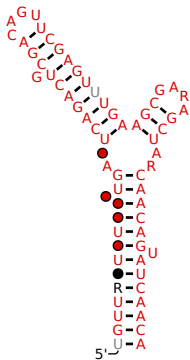
KOKV.1.4.1.1.1.1
NMV.1.4.2.1.1.1
#=GC SS_cons

GUUACCCUGUGUAA CUAAC CAGACAGAU C GAAGUUAGGUGAUUA - CAU AACACAGUGUGAAC
GUUACCCUGCGUGAACUGAUUAGACAGAU C GACAUCAGAUAGUUU CUUUAACUCAGUGUGAAC
(((((((((((((((.....((((.....)).....))))))((((.....))))..))))))..))))))

SLA in different FV groups

Consensus structures computed from mlocarna alignments of all non-redundant representatives for each group obtained from NCBI

Spondweni virus group (SPOVG)

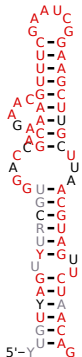


ZIKV.1.6.2.21.1.1	UGUUGAUCUGUGUGAGUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.161.1.1	UGUUGAUCUGUGUGAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.158.1.1	UGUUGAUCUGUGUGAGUCAGACUGCGACAGUUCGAGUCUGAAGCGAGAGCUAAACACAGUAUCAACA
ZIKV.1.6.2.169.1.1	UGUUGAUCUGUGUGAAUCAGACUGCGACAGUUCGAGUUUGAAGCGAGAGCUAAACACAGUAUCAACA
ZIKV.1.6.2.84.1.1	AGUUGAUCUGUGUGAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.150.1.1	UGUU--UCUGUGUGAUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
ZIKV.1.6.2.141.1.1	UGUUACUGUUGCUGACUCAGACUGCGACAGUUCGAGUUUGAAGCGAAAGCUAGCAACAGUAUCAACA
#=GC SS_cons	(((((((((((((.(.(((.(((((((.(.((...)))))))))))))((.(...)).).)))))).))))))

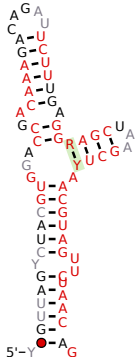
Dengue virus SLA

There are four DENV serotypes, probably five ...

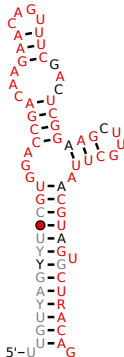
DENV1



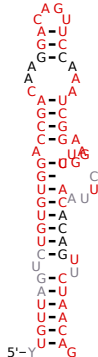
DENV2



DENV3



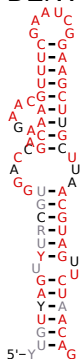
DENV4



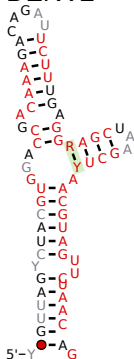
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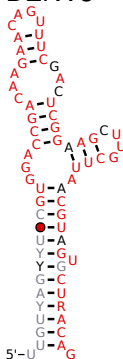
DENV1



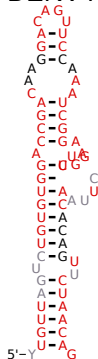
DENV2



DENV3



DENV4



Alternative facts:

- DENV1 folds into another consensus structure
- We need to deconvolute that !

RNAaliSplit objectives

```
DENV1.1.2.1.156.1.1 CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGWUCUAACAG
DENV1.1.2.1.21.1.1 CUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGKUCUAACAG
DENV1.1.2.1.1731.1.1 UUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUCGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.229.1.1 UUGUUAGU-CUGUCUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.339.1.1 UUGUUAGU-CUACGUGGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUACAG
DENV1.1.2.1.157.1.1 UUGUUAGU-CUACGUGGACU-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
DENV1.1.2.1.168.1.1 UUGUUAGUACUACGUGGACC-GACAAGAAUCAGUUUCGAAUCGGAAGCUUGCCGGUUAACGUAGUUCUAACAG
DENV1.1.2.1.402.1.1 GACUCAGA-UACACGCGACC-GACAAGAA-CAGUUUCGAAUCGGAAGCUUGC---UUAACGUAGUUCUAACAG
#=GC SS_cons .((((((.(((((((.(.((...((....)))...)))...)).).....)))))).)))))).
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Structure conservation

Covariation

RNAaliSplit objectives

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How can we quantify them?

RNAaliSplit objectives

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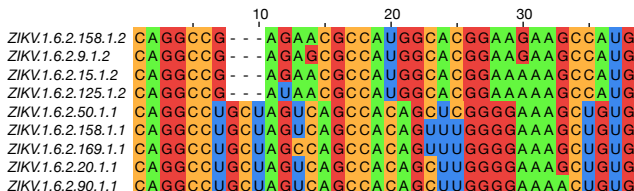
Structure conservation

- SCI
 - z-score
 - mean pairwise identity
 - GC content
- RNAz SVM RNA-class probability

Covariation

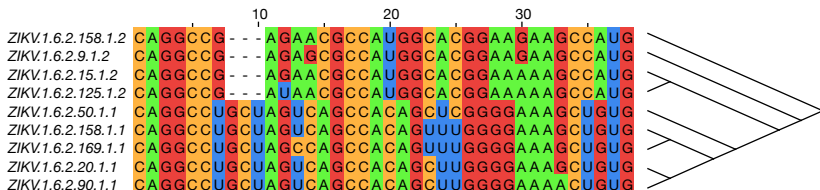
- compensatory mutations
- RNAalifold
- statistical significance
- R-scape

RNAaliSplit workflow



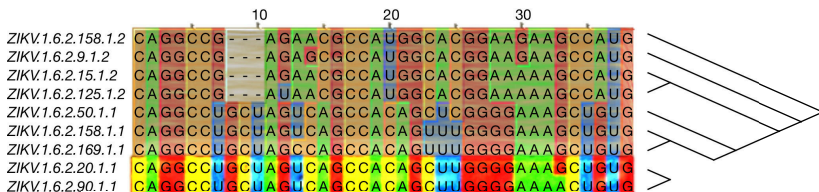
SVM prob	z-score	SCI	seq consensus	structure	SP	alignment
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0.939812	-1.17	0.8497	7	(((((.....(((.....)))))))).))))))	3	split7.set1.aln
0.036973	-0.65	0.9671	2	(((((.....(((.....)))))))).))))))	0	split7.set2.aln
0.051212	-0.51	1.0091	2	(((((.....(((.....)))))))).))))))	0	split8.set1.aln
0.887830	-1.18	0.8409	7	(((((.....(((.....)))))))).))))))	2	split8.set2.aln
0.899553	-1.43	0.9442	5	(((((.....(((.....)))))))).))))))	2	split9.set1.aln
0.063624	-0.56	0.9789	4	(((((.....(((.....)))))))).))))))	0	split9.set2.aln
...						

RNAaliSplit workflow



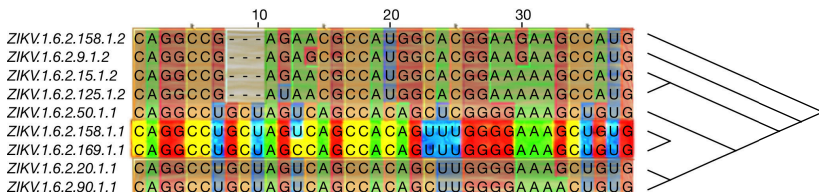
SVM prob	z-score	SCI	seq consensus	structure	SP	alignment
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RNAaliSplit workflow



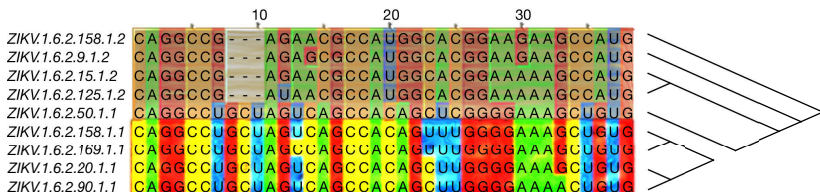
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RNAaliSplit workflow



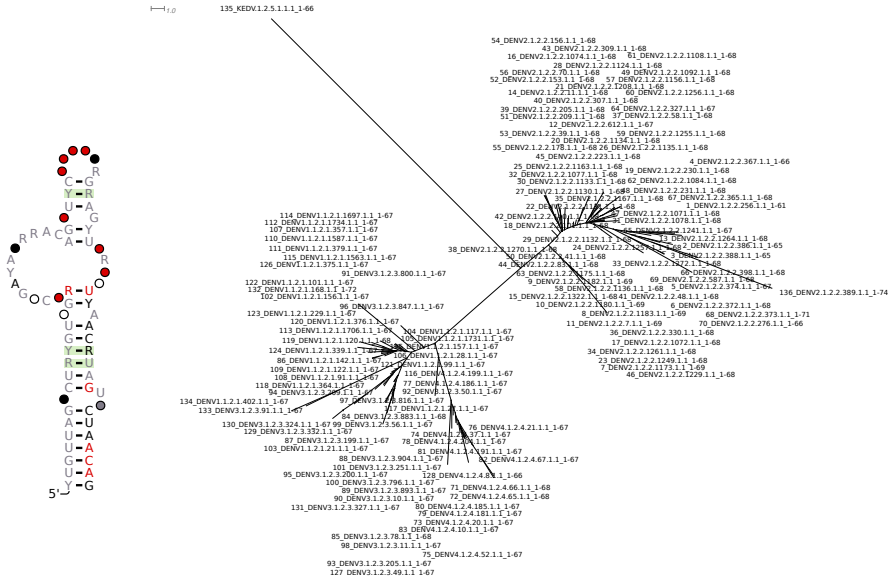
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RNAaliSplit workflow



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...					

Dengue virus SLA



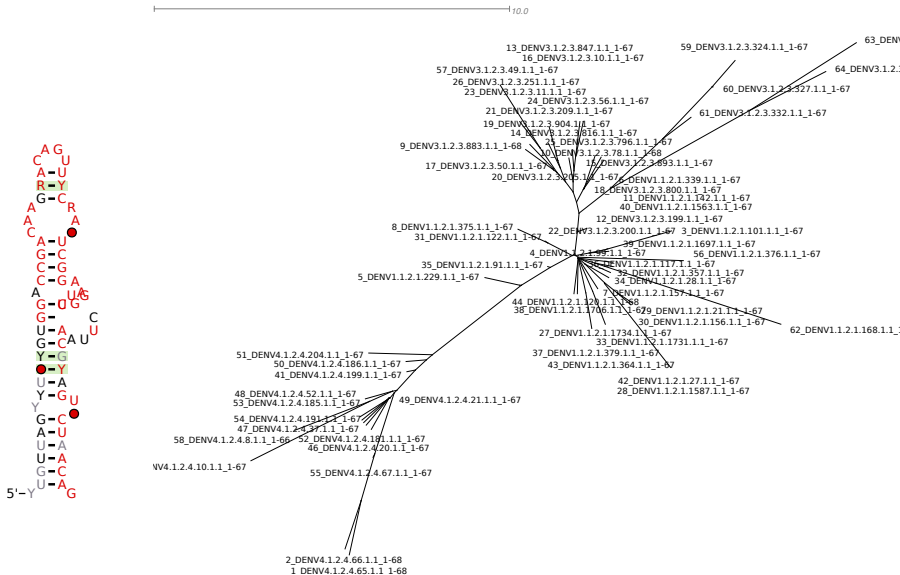
NJ tree; 146 pairwise different DENVG SLA sequences

Dengue virus SLA



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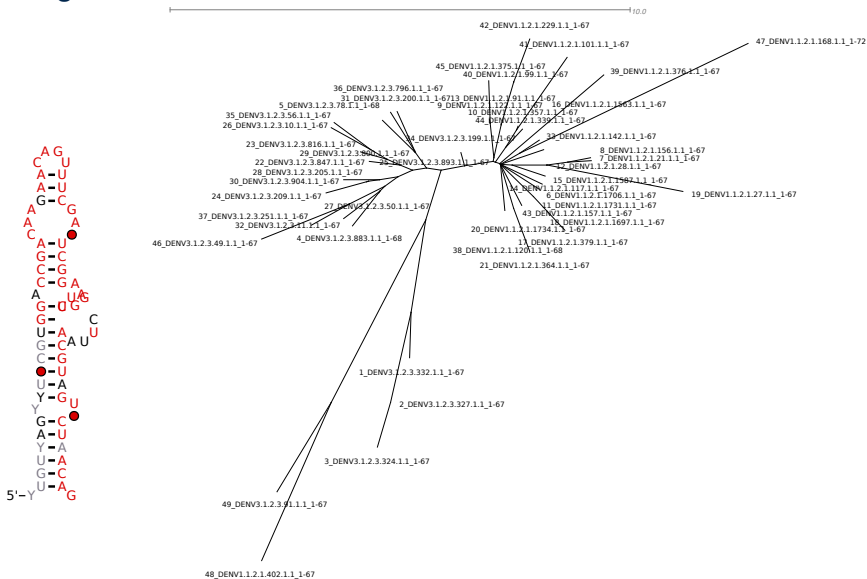
Dengue virus SLA



NJ tree; 64 pairwise different DENV1,3,4 SLA sequences

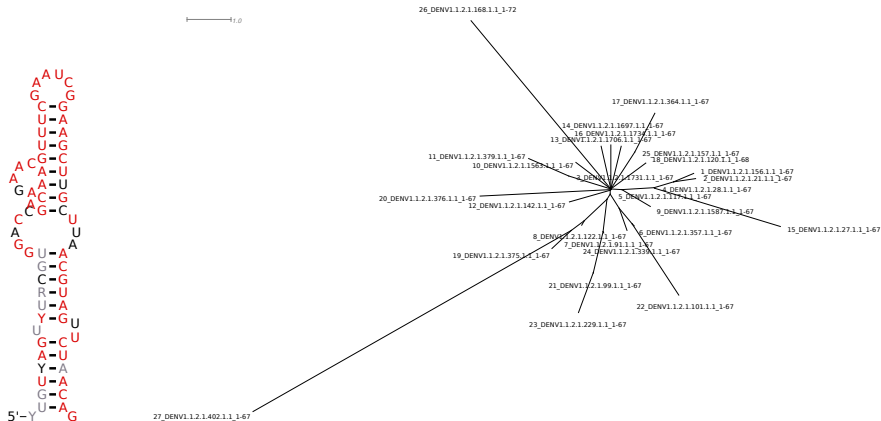


Dengue virus SLA



NJ tree; 49 pairwise different DENV1,3 SLA sequences

Dengue virus SLA

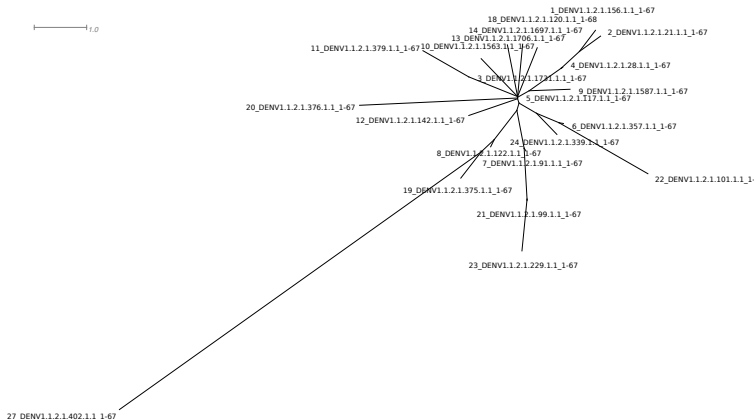
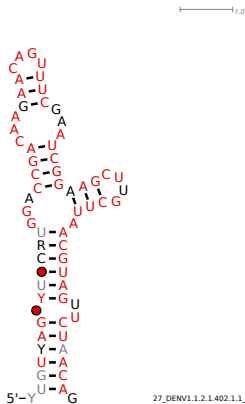


NJ tree; 27 pairwise different DENV1 SLA sequences

5'-Y G



Dengue virus SLA



Summary

Flavivirus SLA elements are

- required for FV replication, mediating (-)gRNA synthesis by RdRp
- conserved among Mosquito-bourne and Tick-bourne FV

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RNAaliSplit is

- a heuristic approach for splitting MSA into sets that have alternative consensus structures
- a means for cleaning MSA before constructing covariance models
- universally applicable to moderate-sized RNA MSA
- implemented in a modular, object-oriented way based on the Perl Moose framework
- available soon as a stand-alone Perl Module

Acknowledgments

Collaborators

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Der Wissenschaftsfonds.

SFB RNA regulation of the
transcriptome (FWF-F43)



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Thank you!